

BRAC University

Department of Mathematics and Physical Sciences

MAT110

Lecture 18

**Reduction of the General Second-Degree
Equation
to Standard Form**

Akram Hosen

Lecturer in Mathematics

Department of Mathematics and Physical Sciences

Contents

1	Reduction to Standard Form	1
1.1	Case I: $ab - h^2 \neq 0$	1
2	The Case $ab - h^2 = 0$	3

1. Reduction to Standard Form

We shall reduce the general equation of the second degree

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

to its standard form.

The Fundamental Quantities

For the general second-degree equation, let

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}, \quad C = ab - h^2.$$

1.1. Case I: $ab - h^2 \neq 0$

We first transfer the origin to the centre (x_1, y_1) of the conic. The equation then becomes

$$ax^2 + 2hxy + by^2 + \frac{\Delta}{C} = 0. \quad (1)$$

Now rotate the axes through the angle

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2h}{a-b} \right)$$

in order to remove the xy term. Equation (1) becomes

$$a'x'^2 + b'y'^2 + \frac{\Delta}{C} = 0. \quad (2)$$

By the theorem of invariants,

$$a' + b' = a + b, \quad a'b' = ab - h^2. \quad (3)$$

Hence a' and b' can be determined from (3). If

$$c' = \frac{\Delta}{C} = \frac{\Delta}{ab - h^2},$$

then (2) may be written as

$$a'x'^2 + b'y'^2 + c' = 0,$$

or

$$\boxed{\frac{x'^2}{-c'/a'} + \frac{y'^2}{-c'/b'} = 1}.$$

This is the standard form of the given equation. It represents either an ellipse or a hyperbola.

Example 1

Reduce the equation

$$x^2 - 4xy + y^2 + 8x + 2y - 5 = 0$$

to its standard form.

Answer

Here,

$$a = 1, \quad b = 1, \quad h = -2, \quad g = 4, \quad f = 1, \quad c = -5.$$

Thus,

$$ab - h^2 = 1 - 4 = -3 < 0.$$

Therefore, the equation represents a hyperbola. Also,

$$\begin{aligned} \Delta &= \begin{vmatrix} 1 & -2 & 4 \\ -2 & 1 & 1 \\ 4 & 1 & -5 \end{vmatrix} \\ &= -18, \end{aligned}$$

and

$$C = ab - h^2 = -3, \quad \frac{\Delta}{C} = \frac{-18}{-3} = 6.$$

Let us now transform the origin to the centre of the conic and then rotate the axes in order to remove the xy term. The final equation is

$$a'x'^2 + b'y'^2 + \frac{\Delta}{C} = 0. \quad (4)$$

By the invariants,

$$a' + b' = a + b = 1 + 1 = 2, \quad a'b' = ab - h^2 = -3.$$

Solving these equations gives

$$a' = -1, \quad b' = 3, \quad \text{or} \quad a' = 3, \quad b' = -1.$$

Hence, from (4),

$$-x'^2 + 3y'^2 + 6 = 0 \implies \boxed{\frac{x'^2}{6} - \frac{y'^2}{2} = 1},$$

or

$$3x'^2 - y'^2 + 6 = 0 \implies \boxed{\frac{y'^2}{6} - \frac{x'^2}{2} = 1}.$$

2. The Case $ab - h^2 = 0$

Put

$$a = \alpha^2, \quad b = \beta^2.$$

Since $ab - h^2 = 0$, we take

$$h = \alpha\beta.$$

Therefore,

$$\begin{aligned} & \alpha x^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \\ \implies & \alpha^2 x^2 + 2\alpha\beta xy + \beta^2 y^2 + 2gx + 2fy + c = 0 \\ \implies & (\alpha x + \beta y)^2 + 2gx + 2fy + c = 0. \end{aligned}$$

Introducing a constant k , write

$$\begin{aligned} (\alpha x + \beta y + k)^2 &= (\alpha x + \beta y)^2 + 2k(\alpha x + \beta y) + k^2 \\ &= -2gx - 2fy + 2k\alpha x + 2k\beta y + k^2 - c. \end{aligned}$$

Thus,

$$(\alpha x + \beta y + k)^2 = 2(k\alpha - g)x + 2(k\beta - f)y + k^2 - c. \quad (5)$$

Choose k such that the two lines

$$\alpha x + \beta y + k = 0 \quad (6)$$

and

$$2(k\alpha - g)x + 2(k\beta - f)y + k^2 - c = 0 \quad (7)$$

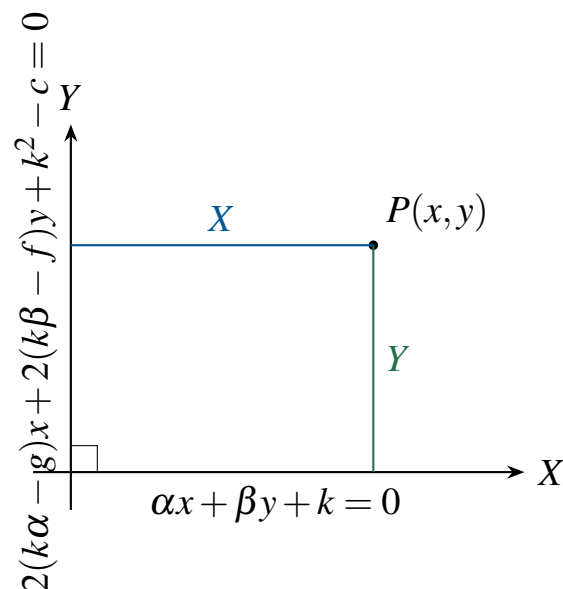
are at right angles. Hence,

$$2\alpha(k\alpha - g) + 2\beta(k\beta - f) = 0,$$

which gives

$$\boxed{k = \frac{\alpha g + \beta f}{\alpha^2 + \beta^2}}. \quad (8)$$

Let the lines (6) and (7) be the new coordinate axes, as shown below.



If a point P on the conic has coordinates x, y referred to the old axes and X, Y referred to the new axes, then

$$Y = \frac{\alpha x + \beta y + k}{\sqrt{\alpha^2 + \beta^2}}, \quad (9)$$

and

$$X = \frac{2(k\alpha - g)x + 2(k\beta - f)y + k^2 - c}{2\sqrt{(k\alpha - g)^2 + (k\beta - f)^2}}. \quad (10)$$

Using (9) and (10) in (5),

$$(\alpha^2 + \beta^2)Y^2 = 2\sqrt{(k\alpha - g)^2 + (k\beta - f)^2}X,$$

or

$$Y^2 = \frac{2\sqrt{(k\alpha - g)^2 + (k\beta - f)^2}}{\alpha^2 + \beta^2}X. \quad (11)$$

Substituting the value of k from (8), equation (11) becomes

$$Y^2 = \frac{2(\alpha f - \beta g)}{(\alpha^2 + \beta^2)^{3/2}}X. \quad (12)$$

Therefore,

$$\boxed{Y^2 = 4AX}, \quad (13)$$

where

$$\boxed{A = \frac{\alpha f - \beta g}{2(\alpha^2 + \beta^2)^{3/2}}}.$$

Thus, the standard form in this case is a parabola.

Example 2

Reduce

$$x^2 + y^2 - 2xy + x + y + 1 = 0$$

to its standard form.

Answer

Here,

$$a = 1, \quad b = 1, \quad h = -1, \quad g = \frac{1}{2}, \quad f = \frac{1}{2}, \quad c = 1.$$

Since

$$ab - h^2 = 1(1) - (-1)^2 = 0,$$

the equation represents a parabola. Now,

$$\begin{aligned} x^2 + y^2 - 2xy + x + y + 1 &= 0 \\ (x - y)^2 + x + y + 1 &= 0 \\ (x - y)^2 &= -x - y - 1. \end{aligned} \tag{14}$$

Introducing k ,

$$\begin{aligned} (x - y + k)^2 &= (x - y)^2 + 2k(x - y) + k^2 \\ &= -x - y - 1 + 2kx - 2ky + k^2 \\ &= (2k - 1)x - (2k + 1)y + k^2 - 1. \end{aligned} \tag{15}$$

Determine k such that the lines

$$x - y + k = 0$$

and

$$(2k - 1)x - (2k + 1)y + k^2 - 1 = 0$$

are at right angles. Therefore,

$$1(2k - 1) + (-1)(-2k - 1) = 0,$$

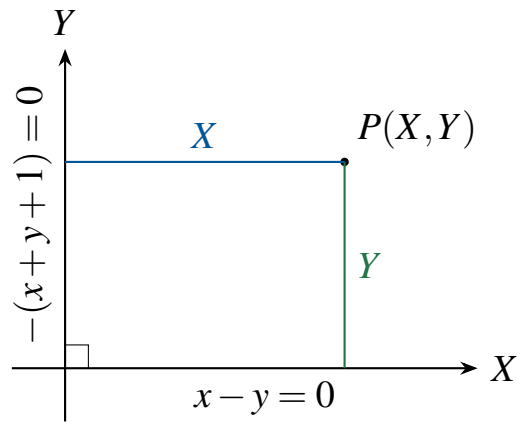
so

$$k = 0.$$

Equation (15) becomes

$$(x - y)^2 = -(x + y + 1). \tag{16}$$

Take the lines $x - y = 0$ and $-(x + y + 1) = 0$ as the new X - and Y -axes, respectively.



The new coordinates are

$$Y = \frac{x-y}{\sqrt{2}}, \quad X = \frac{-(x+y+1)}{\sqrt{2}}.$$

Hence,

$$x-y = \sqrt{2}Y, \quad -(x+y+1) = \sqrt{2}X.$$

Using these relations in (16),

$$(\sqrt{2}Y)^2 = \sqrt{2}X,$$

so

$$\sqrt{2}Y^2 = X.$$

Therefore, the required standard form is

$$Y^2 = \frac{1}{\sqrt{2}}X.$$